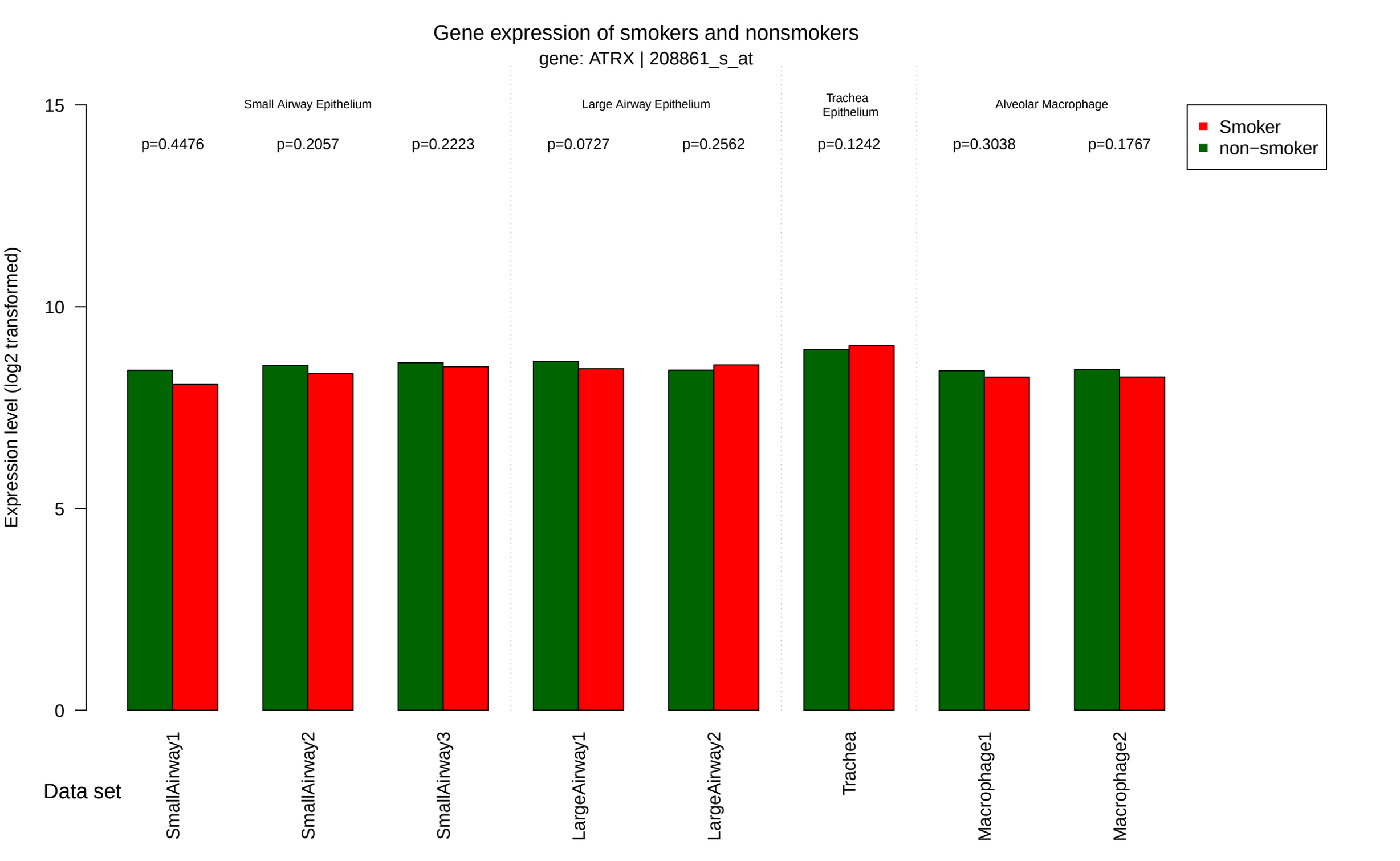
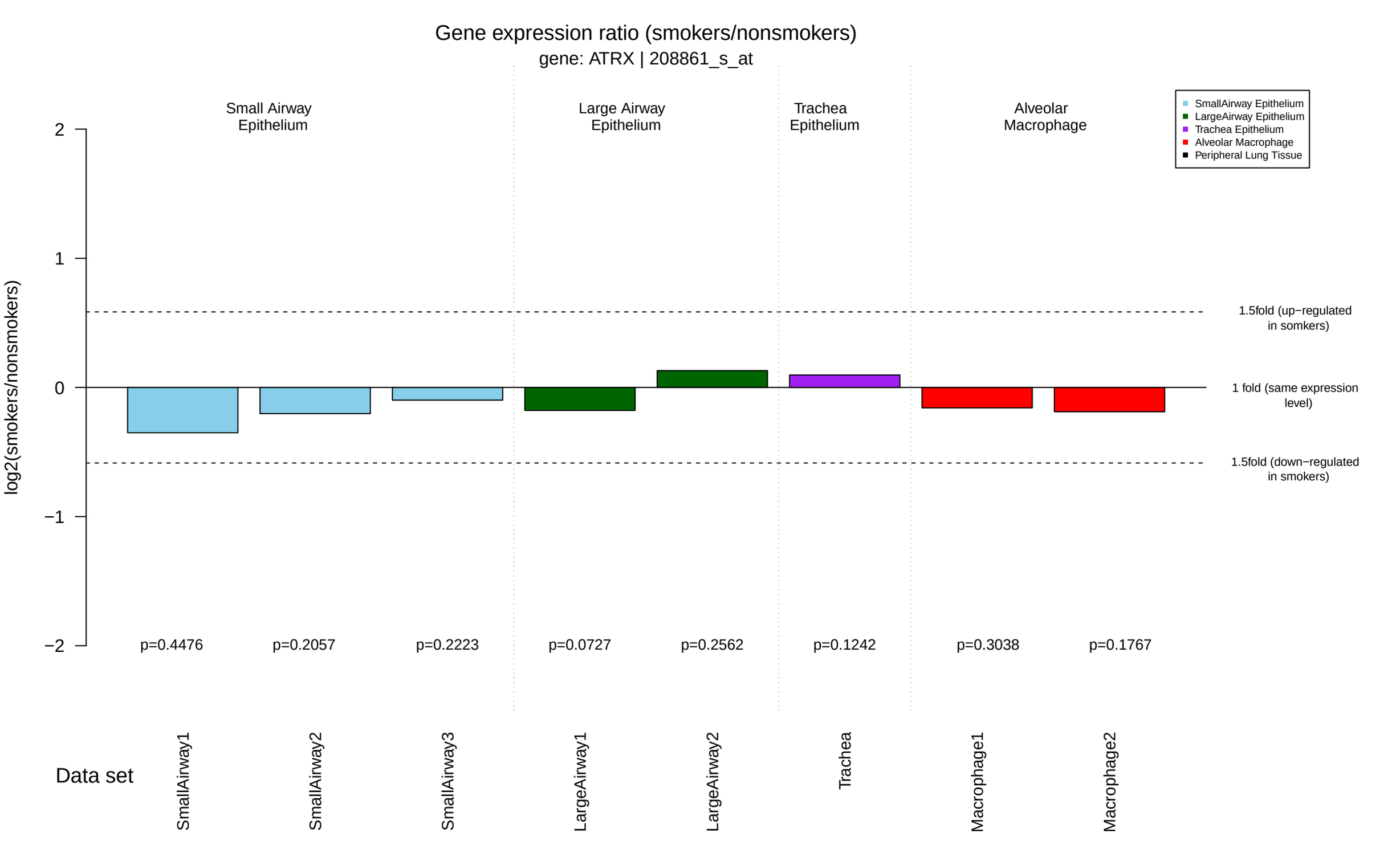
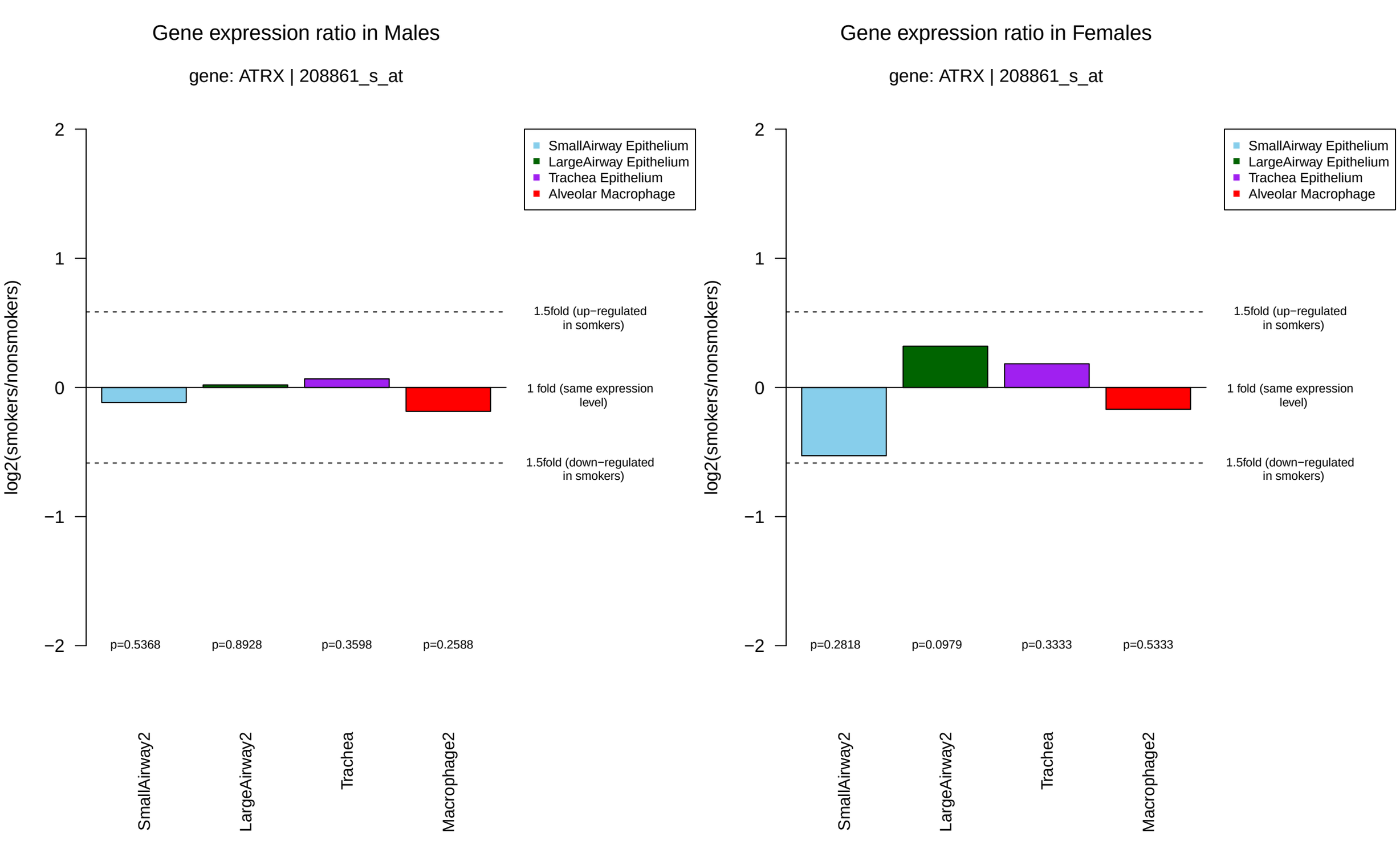
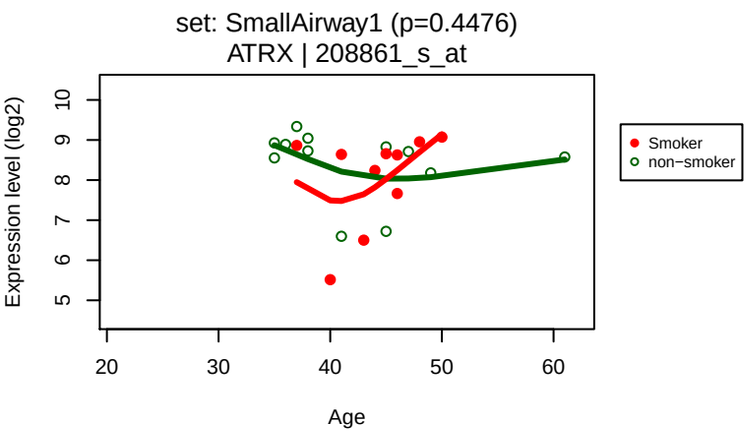
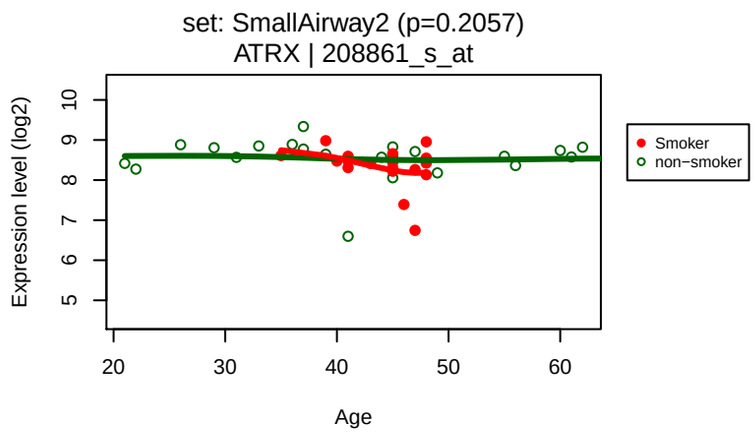
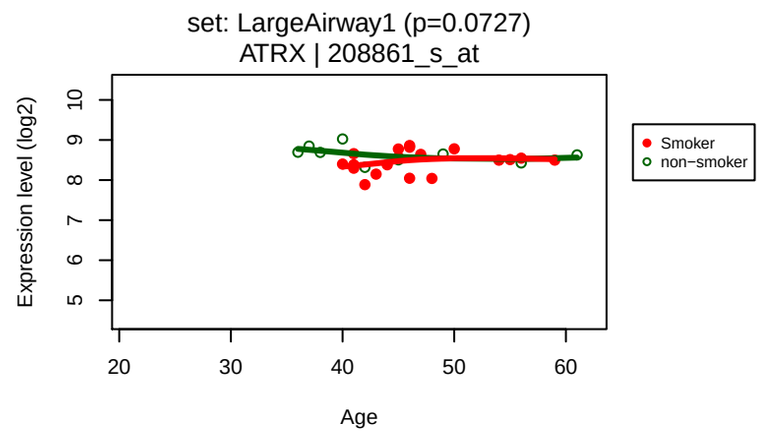

Gene expression of smokers and nonsmokers
gene: ATRX | 208861_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4476
p=0.2057
p=0.2223
p=0.0727
p=0.2562
p=0.1242
p=0.3038
p=0.1767
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

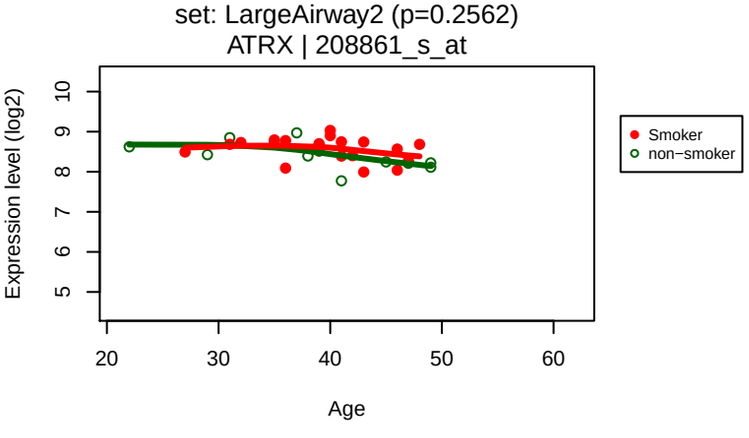

Gene expression ratio (smokers/nonsmokers)
gene: ATRX | 208861_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4476
p=0.2057
p=0.2223
p=0.0727
p=0.2562
p=0.1242
p=0.3038
p=0.1767
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

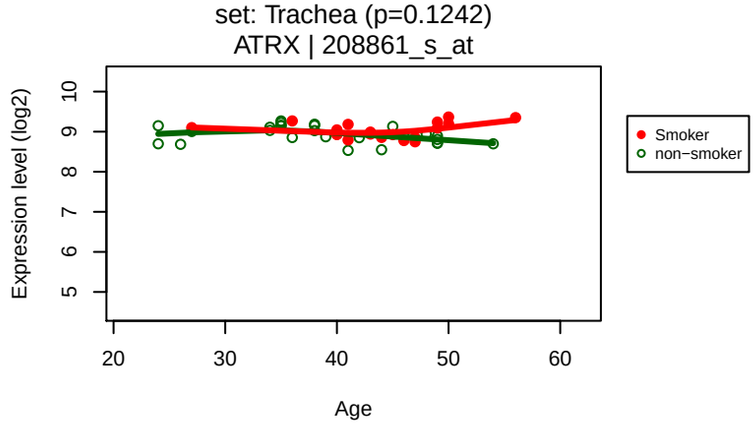

Gene expression ratio in Males
gene: ATRX | 208861_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5368
p=0.8928
p=0.3598
p=0.2588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATRX | 208861_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2818
p=0.0979
p=0.3333
p=0.5333
SmallAirway2
LargeAirway2
Trachea
Macrophage2

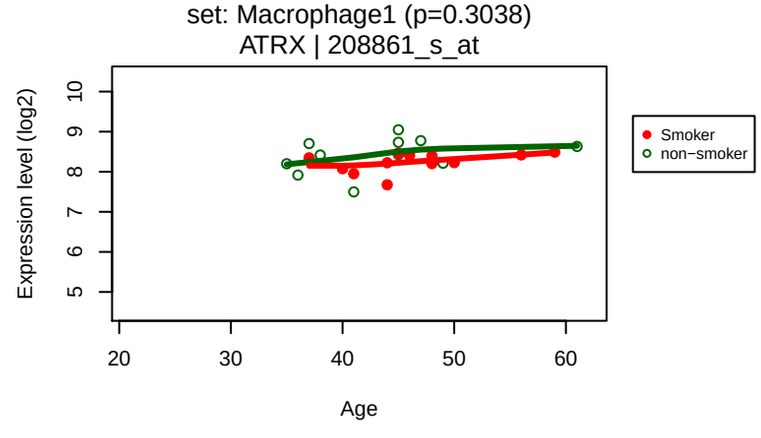

set: SmallAirway1 (p=0.4476)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker

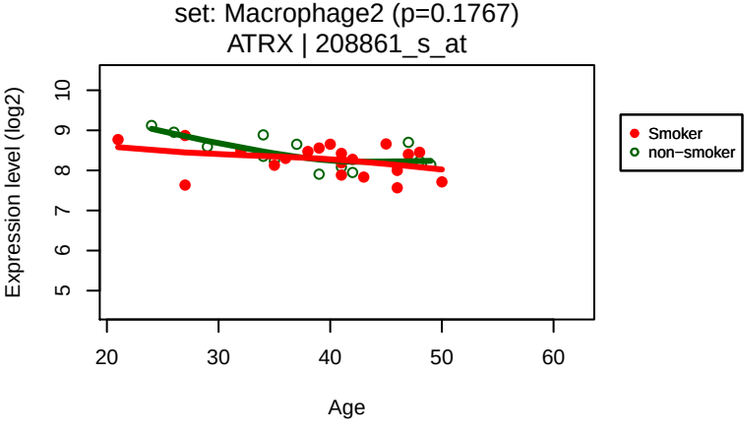

set: SmallAirway2 (p=0.2057)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker

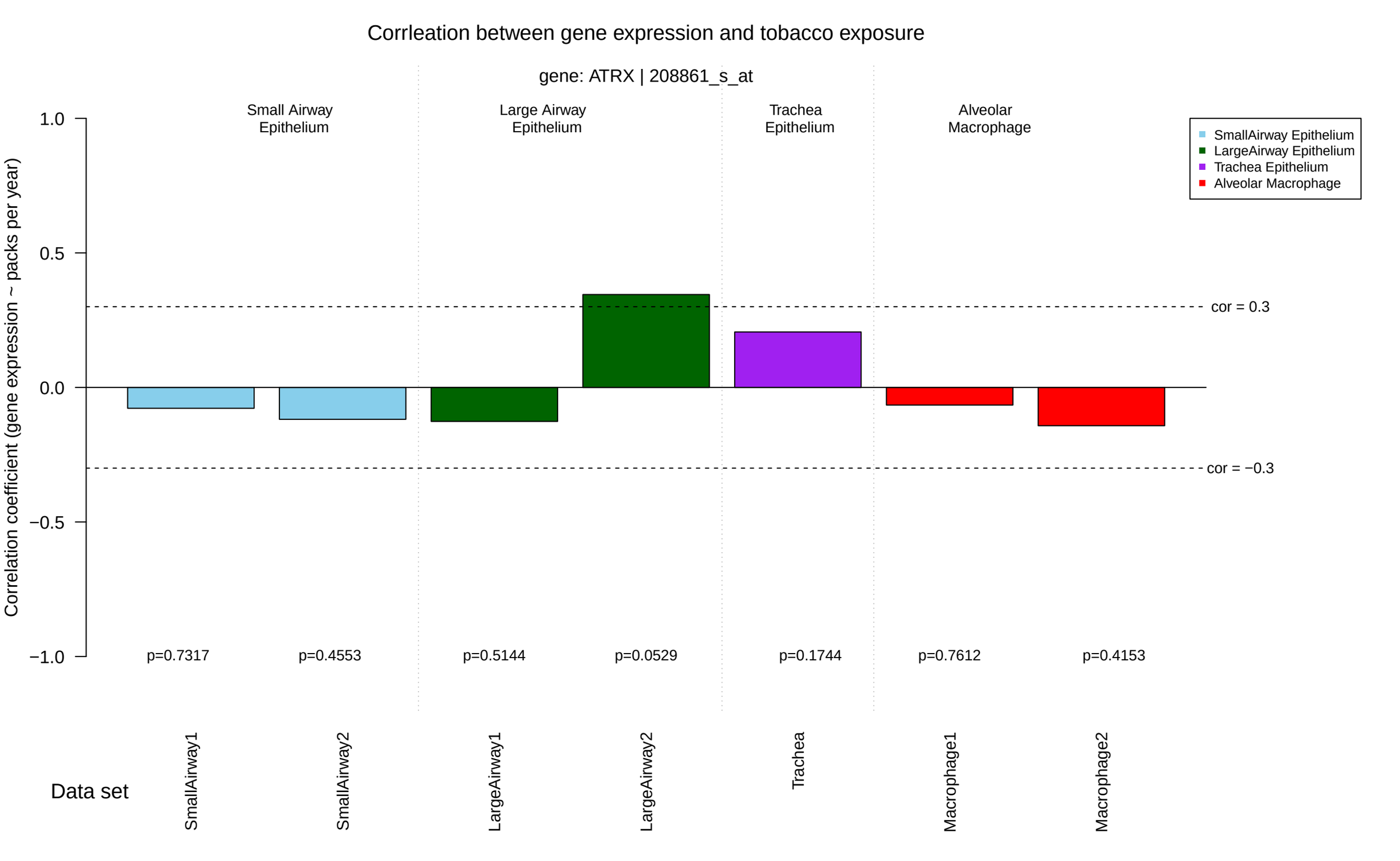

set: LargeAirway1 (p=0.0727)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2562)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1242)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3038)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1767)
ATRX | 208861_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATRX | 208861_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7317
p=0.4553
p=0.5144
p=0.0529
p=0.1744
p=0.7612
p=0.4153
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2